

KRASNOYARSK, Russian Federation

WMO: 295740

Lat: **56.000N**Lon: **92.883E**Elev: **277**StdP: **98.04**Time Zone: **7.00 (E07)**Period: **82-94**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.7	-31.1	-36.1	0.1	-33.5	-33.4	0.2	-30.9	12.6	-11.1	11.6	-11.0	1.3	290	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	28.4	18.3	26.6	17.6	24.8	16.8	20.0	25.7	19.0	24.3	18.0	23.0	2.9	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	13.4	22.4	17.0	12.6	21.3	16.1	11.8	20.5	58.9	25.7	55.3	24.4	52.3	23.2	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.1	8.4	7.0	-35.3	31.6	4.0	2.1	-38.2	33.1	-40.5	34.3	-42.7	35.4	-45.6	36.9		
			-34.9	22.1	4.1	2.0	-37.8	23.5	-40.3	24.7	-42.6	25.9	-45.6	27.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.4	-14.6	-13.5	-6.6	1.3	9.5	15.2	18.4	15.6	8.8	2.0	-7.3	-13.1
	DBStd	13.03	8.35	7.44	6.13	5.59	5.26	4.23	3.17	2.97	3.84	4.89	7.12	8.03
	HDD10.0	3837	762	657	516	263	76	8	0	1	68	251	519	716
	HDD18.3	6254	1021	890	774	512	278	111	38	93	287	507	769	974
	CDD10.0	694	0	0	0	2	59	162	262	175	31	2	0	0
	CDD18.3	68	0	0	0	0	3	16	41	8	0	0	0	0
	CDH23.3	682	0	0	0	0	56	184	363	74	4	0	0	0
	CDH26.7	137	0	0	0	0	13	34	83	8	0	0	0	0
Wind	WSAvg	2.8	2.7	2.7	2.8	3.7	3.3	2.5	2.0	2.0	2.2	3.3	3.5	3.2
Precipitation	PrecAvg	496	17	14	17	31	46	62	76	73	45	43	41	29
	PrecMax	665	35	40	48	73	99	147	195	151	112	142	119	71
	PrecMin	322	2	3	2	2	4	18	7	14	13	6	5	5
	PrecStd	84	8	8	10	15	24	27	34	33	22	24	22	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.0	3.7	9.8	18.2	28.5	29.6	31.1	27.7	23.5	17.5	6.2	3.3
		MCWB	-1.5	-0.3	4.5	8.7	14.7	18.4	20.3	18.6	15.0	9.3	2.9	-0.4
	2%	DB	-0.3	0.5	6.0	15.1	24.7	27.7	29.0	25.5	20.4	13.9	4.4	0.7
		MCWB	-2.7	-2.7	1.5	7.0	13.4	17.5	19.3	17.2	13.9	7.8	1.0	-2.2
	5%	DB	-1.9	-1.5	3.7	12.2	22.0	25.7	27.2	23.8	17.6	11.5	2.8	-1.4
		MCWB	-3.8	-3.9	0.2	6.0	12.2	16.6	18.9	16.5	12.3	6.8	0.1	-3.5
	10%	DB	-3.4	-3.8	1.7	9.5	19.1	23.5	25.3	22.0	15.5	9.0	1.3	-3.1
		MCWB	-5.2	-5.8	-1.0	4.5	10.9	15.4	18.2	15.9	11.2	5.2	-0.8	-4.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.4	-0.1	4.5	10.2	16.1	20.3	22.6	19.7	16.5	10.3	4.3	0.5
		MCDB	1.7	3.2	9.5	14.7	24.6	27.8	27.8	23.9	21.2	15.7	5.5	2.4
	2%	WB	-2.4	-2.4	1.8	8.0	14.3	18.9	20.6	18.6	14.2	8.7	1.6	-2.0
		MCDB	-0.5	0.5	5.2	13.8	22.7	25.4	26.2	23.3	18.7	12.6	3.4	0.6
	5%	WB	-3.8	-3.9	0.4	6.2	12.9	17.6	19.7	17.6	12.9	7.3	0.5	-3.3
		MCDB	-1.9	-1.6	3.3	11.9	20.3	23.5	25.1	22.0	17.0	11.2	2.5	-1.4
	10%	WB	-5.2	-5.7	-0.8	4.6	11.7	16.4	18.8	16.6	11.7	5.7	-0.8	-4.7
		MCDB	-3.4	-3.9	1.6	9.1	18.2	21.8	23.8	20.7	14.9	8.6	1.2	-3.1
Mean Daily Temperature Range	5% DB	MDBR	7.9	8.9	9.5	9.3	10.7	11.0	10.3	10.1	9.6	7.3	6.7	7.3
		MCDBR	7.8	9.4	10.9	13.7	15.8	14.9	13.7	13.6	14.2	11.7	7.2	8.9
		MCWBR	6.3	7.4	7.4	7.8	7.6	6.5	5.8	6.7	8.0	7.1	5.5	7.1
	5% WB	MCDBR	7.6	9.1	10.3	12.8	13.6	12.5	11.4	10.9	12.4	10.6	7.6	8.7
		MCWBR	6.4	7.4	7.3	7.8	6.9	6.2	5.7	6.5	7.5	7.0	6.5	7.3
Clear-Sky Solar Irradiance	taub	0.281	0.314	0.361	0.394	0.411	0.407	0.429	0.410	0.368	0.349	0.316	0.283	
	taud	2.031	2.014	1.979	2.141	2.198	2.307	2.258	2.277	2.346	2.246	2.072	1.977	
	Ebn at Noon	585	716	775	818	828	838	807	796	781	683	529	463	
	Edh at Noon	65	99	132	131	132	120	124	114	93	79	63	54	
All-Sky Solar Radiation	RadAvg	0.80	1.79	3.14	4.29	4.93	5.93	5.28	4.24	2.73	1.45	0.72	0.48	
	RadStd	0.09	0.12	0.24	0.28	0.53	0.49	0.36	0.38	0.36	0.12	0.10	0.06	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (19 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page